Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
use the properties of angles.			
use the properties of triangles.			
use the Pythagorean Theorem.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

9.4 Use Properties of Rectangles, Triangles, and Trapezoids

Learning Objectives

By the end of this section, you will be able to:

- Understand linear, square, and cubic measure
- Use properties of rectangles
- Use properties of triangles
- Use properties of trapezoids

☑ **BE PREPARED** 9.10 Before you get started, take this readiness quiz.

The length of a rectangle is 3 less than the width. Let w represent the width. Write an expression for the length of the rectangle.
If you missed this problem, review Example 2.26.

☑ **BE PREPARED** 9.11 Simplify: $\frac{1}{2}(6h)$.
If you missed this problem, review Example 7.7.

☑ **BE PREPARED** 9.12 Simplify: $\frac{5}{2}(10.3-7.9)$.
If you missed this problem, review Example 5.36.

In this section, we'll continue working with geometry applications. We will add some more properties of triangles, and we'll learn about the properties of rectangles and trapezoids.

Understand Linear, Square, and Cubic Measure

When you measure your height or the length of a garden hose, you use a ruler or tape measure (Figure 9.13). A tape measure might remind you of a line—you use it for **linear measure**, which measures length. Inch, foot, yard, mile, centimeter and meter are units of linear measure.



Figure 9.13 This tape measure measures inches along the top and centimeters along the bottom.

When you want to know how much tile is needed to cover a floor, or the size of a wall to be painted, you need to know the **area**, a measure of the region needed to cover a surface. Area is measured is **square units**. We often use square inches, square feet, square centimeters, or square miles to measure area. A square centimeter is a square that is one centimeter (cm) on each side. A square inch is a square that is one inch on each side (Figure 9.14).

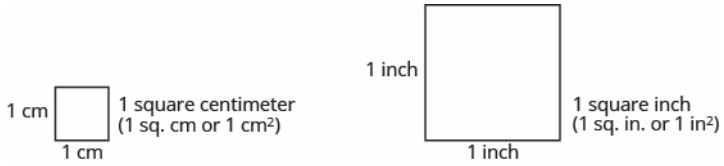


Figure 9.14 Square measures have sides that are each 1 unit in length.

Figure 9.15 shows a rectangular rug that is 2 feet long by 3 feet wide. Each square is 1 foot wide by 1 foot long, or 1 square foot. The rug is made of 6 squares. The area of the rug is6 square feet.

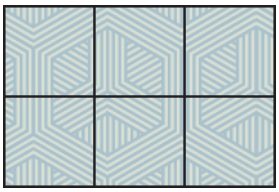


Figure 9.15 The rug contains six squares of 1 square foot each, so the total area of the rug is 6 square feet.

When you measure how much it takes to fill a container, such as the amount of gasoline that can fit in a tank, or the amount of medicine in a syringe, you are measuring **volume**. Volume is measured in **cubic units** such as cubic inches or cubic centimeters. When measuring the volume of a rectangular solid, you measure how many cubes fill the container. We often use cubic centimeters, cubic inches, and cubic feet. A cubic centimeter is a cube that measures one centimeter on each side, while a cubic inch is a cube that measures one inch on each side (Figure 9.16).

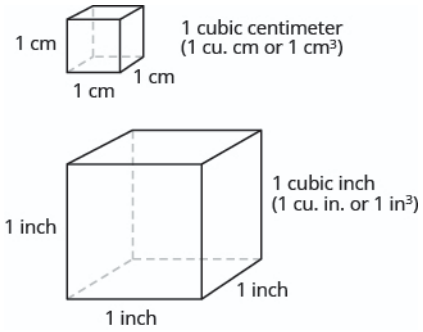


Figure 9.16 Cubic measures have sides that are 1 unit in length.

Suppose the cube in Figure 9.17 measures 3 inches on each side and is cut on the lines shown. How many little cubes does it contain? If we were to take the big cube apart, we would find 27 little cubes, with each one measuring one inch on all sides. So each little cube has a volume of 1 cubic inch, and the volume of the big cube is 27 cubic inches.

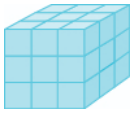


Figure 9.17 A cube that measures 3 inches on each side is made up of 27 one-inch cubes, or 27 cubic inches.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity Visualizing Area and Perimeter will help you develop a better understanding of the difference between the area of a figure and its perimeter.

EXAMPLE 9.25

For each item, state whether you would use linear, square, or cubic measure:

ⓐ amount of carpeting needed in a room ⓑ extension cord length ⓒ amount of sand in a sandbox

ⓓ length of a curtain rod ⓔ amount of flour in a canister ⓕ size of the roof of a doghouse.

Solution

ⓐ You are measuring how much surface the carpet covers, which is the area.	square measure
ⓑ You are measuring how long the extension cord is, which is the length.	linear measure

ⓒ You are measuring the volume of the sand.	cubic measure
ⓓ You are measuring the length of the curtain rod.	linear measure
ⓔ You are measuring the volume of the flour.	cubic measure
ⓕ You are measuring the area of the roof.	square measure

TRY IT 9.49 Determine whether you would use linear, square, or cubic measure for each item.

ⓐ amount of paint in a can ⓑ height of a tree ⓒ floor of your bedroom ⓓ diameter of bike wheel ⓔ size of a piece of sod ⓕ amount of water in a swimming pool

TRY IT 9.50 Determine whether you would use linear, square, or cubic measure for each item.

ⓐ volume of a packing box ⓑ size of patio ⓒ amount of medicine in a syringe ⓓ length of a piece of yarn ⓔ size of housing lot ⓕ height of a flagpole

Many geometry applications will involve finding the perimeter or the area of a figure. There are also many applications of perimeter and area in everyday life, so it is important to make sure you understand what they each mean.

Picture a room that needs new floor tiles. The tiles come in squares that are a foot on each side—one square foot. How many of those squares are needed to cover the floor? This is the area of the floor.

Next, think about putting new baseboard around the room, once the tiles have been laid. To figure out how many strips are needed, you must know the distance around the room. You would use a tape measure to measure the number of feet around the room. This distance is the perimeter.

Perimeter and Area

The **perimeter** is a measure of the distance around a figure.

The **area** is a measure of the surface covered by a figure.

Figure 9.18 shows a square tile that is 1 inch on each side. If an ant walked around the edge of the tile, it would walk 4 inches. This distance is the perimeter of the tile.

Since the tile is a square that is 1 inch on each side, its area is one square inch. The area of a shape is measured by determining how many square units cover the shape.

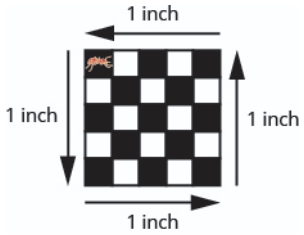


Figure 9.18 $\text{Perimeter} = 4 \text{ inches}$
$\text{Area} = 1 \text{ square inch}$
When the ant walks completely around the tile on its edge, it is tracing the perimeter of the tile. The area of the tile is 1 square inch.

MANIPULATIVE MATHEMATICS

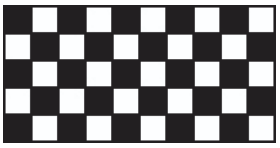
Doing the Manipulative Mathematics activity Measuring Area and Perimeter will help you develop a better understanding of how to measure the area and perimeter of a figure.

EXAMPLE 9.26

Each of two square tiles is 1 square inch. Two tiles are shown together.

ⓐ What is the perimeter of the figure?

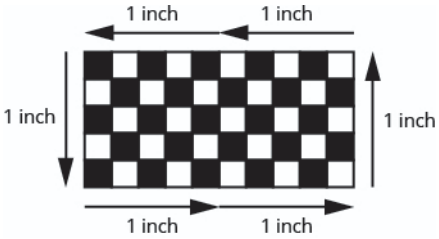
ⓑ What is the area?



Solution

ⓐ The perimeter is the distance around the figure. The perimeter is 6 inches.

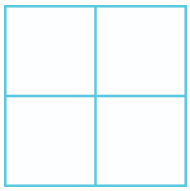
ⓑ The area is the surface covered by the figure. There are 2 square inch tiles so the area is 2 square inches.



TRY IT 9.51 Each box in the figure below is 1 square inch. Find the ⓐ perimeter and ⓑ area of the figure:



TRY IT 9.52 Each box in the figure below is 1 square inch. Find the ⓐ perimeter and ⓑ area of the figure:



Use the Properties of Rectangles

A **rectangle** has four sides and four right angles. The opposite sides of a rectangle are the same length. We refer to one side of the rectangle as the length, L, and the adjacent side as the width, W. See Figure 9.19.

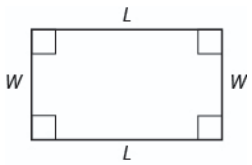


Figure 9.19 A rectangle has four sides, and four right angles. The sides are labeled L for length and W for width.

The perimeter, P, of the rectangle is the distance around the rectangle. If you started at one corner and walked around

the rectangle, you would walk $L + W + L + W$ units, or two lengths and two widths. The perimeter then is

$$P = L + W + L + W$$

or

$$P = 2L + 2W$$

What about the area of a rectangle? Remember the rectangular rug from the beginning of this section. It was 2 feet long by 3 feet wide, and its area was 6 square feet. See Figure 9.20. Since $A = 2 \cdot 3$, we see that the area, A, is the length, L, times the width, W, so the area of a rectangle is $A = L \cdot W$.

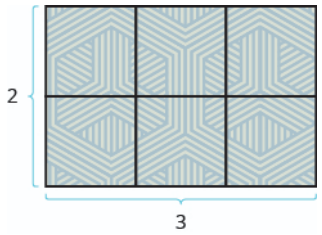


Figure 9.20 The area of this rectangular rug is 6 square feet, its length times its width.

Properties of Rectangles

- Rectangles have four sides and four right $(90°)$ angles.
- The lengths of opposite sides are equal.
- The perimeter, P, of a rectangle is the sum of twice the length and twice the width. See Figure 9.19.
$$P = 2L + 2W$$
- The area, A, of a rectangle is the length times the width.
$$A = L \cdot W$$

For easy reference as we work the examples in this section, we will restate the Problem Solving Strategy for Geometry Applications here.

HOW TO

Use a Problem Solving Strategy for Geometry Applications

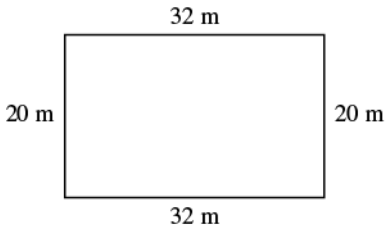
Step 1. **Read** the problem and make sure you understand all the words and ideas. Draw the figure and label it with the given information.
Step 2. **Identify** what you are looking for.
Step 3. **Name** what you are looking for. Choose a variable to represent that quantity.
Step 4. **Translate** into an equation by writing the appropriate formula or model for the situation. Substitute in the given information.
Step 5. **Solve** the equation using good algebra techniques.
Step 6. **Check** the answer in the problem and make sure it makes sense.
Step 7. **Answer** the question with a complete sentence.

EXAMPLE 9.27

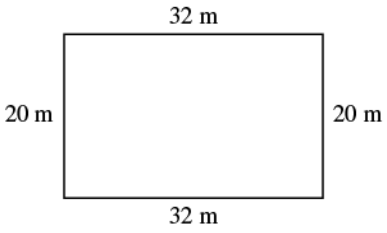
The length of a rectangle is 32 meters and the width is 20 meters. Find (a) the perimeter, and (b) the area.

Solution

ⓐ

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the perimeter of a rectangle
Step 3. **Name.** Choose a variable to represent it.	Let P = the perimeter
Step 4. **Translate.** Write the appropriate formula. Substitute.	$\underbrace{P}\ \underbrace{=}\ \underbrace{2\,L}\ \underbrace{+}\ \underbrace{2\,W}$ $P = 2(32) + 2(20)$
Step 5. **Solve** the equation.	$P = 64 + 40$ $P = 104$
Step 6. **Check:** $P \stackrel{?}{=} 104$ $20 + 32 + 20 + 32 \stackrel{?}{=} 104$ $104 = 104$ ✓	
Step 7. **Answer** the question.	The perimeter of the rectangle is 104 meters.

ⓑ

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the area of a rectangle
Step 3. **Name.** Choose a variable to represent it.	Let A = the area
Step 4. **Translate.** Write the appropriate formula. Substitute.	$\underbrace{A}\ \underbrace{=}\ \underbrace{L}\ \underbrace{\cdot}\ \underbrace{W}$ $A = 32\text{ m} \cdot 20\text{ m}$

Step 5. **Solve** the equation.	$A = 640$
Step 6. **Check:** $A \stackrel{?}{=} 640$ $32 \cdot 20 \stackrel{?}{=} 640$ $640 = 640$ ✓	
Step 7. **Answer** the question.	The area of the rectangle is 640 square meters.

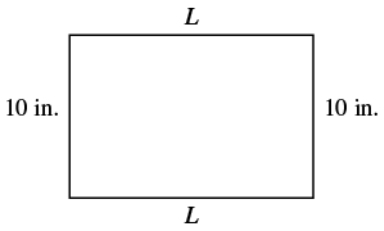
TRY IT 9.53 The length of a rectangle is 120 yards and the width is 50 yards. Find (a) the perimeter and (b) the area.

TRY IT 9.54 The length of a rectangle is 62 feet and the width is 48 feet. Find (a) the perimeter and (b) the area.

EXAMPLE 9.28

Find the length of a rectangle with perimeter 50 inches and width 10 inches.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the length of the rectangle
Step 3. **Name.** Choose a variable to represent it.	Let L = the length
Step 4. **Translate.** Write the appropriate formula. Substitute.	$\underbrace{P}_{50} \quad \underbrace{=}_{=} \quad \underbrace{2\,L}_{2L} \quad \underbrace{+}_{+} \quad \underbrace{2\,W}_{2(10)}$
Step 5. **Solve** the equation.	$50 - 20 = 2L + 20 - 20$ $30 = 2L$ $\frac{30}{2} = \frac{2L}{2}$ $15 = L$

Step 6. **Check:**

$$P = 50$$
$$15 + 10 + 15 + 10 \stackrel{?}{=} 50$$
$$50 = 50 \checkmark$$

Step 7. **Answer** the question.	The length is 15 inches.

TRY IT 9.55 Find the length of a rectangle with a perimeter of 80 inches and width of 25 inches.

TRY IT 9.56 Find the length of a rectangle with a perimeter of 30 yards and width of 6 yards.

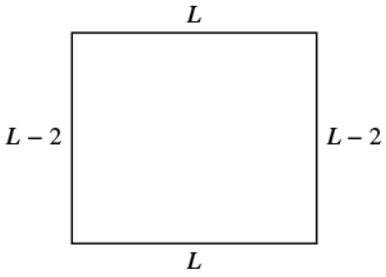
In the next example, the width is defined in terms of the length. We'll wait to draw the figure until we write an expression for the width so that we can label one side with that expression.

EXAMPLE 9.29

The width of a rectangle is two inches less than the length. The perimeter is 52 inches. Find the length and width.

Solution

Step 1. **Read** the problem.

Step 2. **Identify** what you are looking for.	the length and width of the rectangle
Step 3. **Name.** Choose a variable to represent it. Now we can draw a figure using these expressions for the length and width.	Since the width is defined in terms of the length, we let L = length. The width is two feet less that the length, so we let $L - 2$ = width 
Step 4.**Translate.** Write the appropriate formula. The formula for the perimeter of a rectangle relates all the information. Substitute in the given information.	$\underbrace{P}_{52} \quad \underbrace{=}_{=} \quad \underbrace{2L}_{2L} \quad \underbrace{+}_{+} \quad \underbrace{2W}_{2(L-2)}$
Step 5. **Solve** the equation.	$52 = 2L + 2L - 4$
Combine like terms.	$52 = 4L\text{-}4$
Add 4 to each side.	$56 = 4L$
Divide by 4.	$\frac{56}{4} = \frac{4L}{4}$

	$14 = L$
	The length is 14 inches.
Now we need to find the width.	
The width is $L - 2$.	$L - 2$ $14 - 2$ 12 The width is 12 inches.
Step 6. **Check:** Since $14 + 12 + 14 + 12 = 52$, this works!	
Step 7. **Answer** the question.	The length is 14 feet and the width is 12 feet.

TRY IT 9.57 The width of a rectangle is seven meters less than the length. The perimeter is 58 meters. Find the length and width.

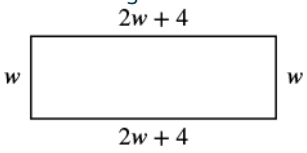
TRY IT 9.58 The length of a rectangle is eight feet more than the width. The perimeter is 60 feet. Find the length and width.

EXAMPLE 9.30

The length of a rectangle is four centimeters more than twice the width. The perimeter is 32 centimeters. Find the length and width.

Solution

Step 1. **Read** the problem.

Step 2. **Identify** what you are looking for.	the length and width
Step 3. **Name.** Choose a variable to represent it.	let W = width The length is four more than twice the width. $2w + 4$ = length 
Step 4.**Translate.** Write the appropriate formula and substitute in the given information.	$\underbrace{P}_{32} \quad \underbrace{=}_{=} \quad \underbrace{2L}_{2(2w+4)} \quad \underbrace{+}_{+} \quad \underbrace{2W}_{2w}$

Step 5. **Solve** the equation.	$32 = 4w + 8 + 2w$ $32 = 6w + 8$ $24 = 6w$ $4 = w$ width $2w + 4$ length $2(4) + 4$ 12 The length is 12 cm.
Step 6. **Check:** $p = 2L + 2W$ $32 \stackrel{?}{=} 2 \cdot 12 + 2 \cdot 4$ $32 = 32$	
Step 7. **Answer** the question.	The length is 12 cm and the width is 4 cm.

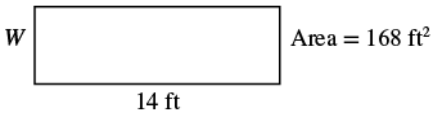
TRY IT 9.59 The length of a rectangle is eight more than twice the width. The perimeter is 64 feet. Find the length and width.

TRY IT 9.60 The width of a rectangle is six less than twice the length. The perimeter is 18 centimeters. Find the length and width.

EXAMPLE 9.31

The area of a rectangular room is 168 square feet. The length is 14 feet. What is the width?

Solution

Step 1. **Read** the problem.	
Step 2. **Identify** what you are looking for.	the width of a rectangular room
Step 3. **Name.** Choose a variable to represent it.	Let W = width
Step 4.**Translate.** Write the appropriate formula and substitute in the given information.	$A = LW$ $168 = 14W$
Step 5. **Solve** the equation.	$\frac{168}{14} = \frac{14W}{14}$ $12 = W$

Step 6. **Check:**

$A = LW$

$168 \stackrel{?}{=} 14 \cdot 12$

$168 = 168$ ✓

Step 7. **Answer** the question.	The width of the room is 12 feet.

TRY IT 9.61 The area of a rectangle is 598 square feet. The length is 23 feet. What is the width?

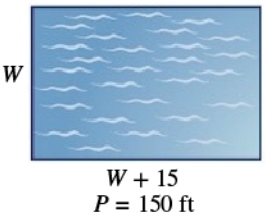
TRY IT 9.62 The width of a rectangle is 21 meters. The area is 609 square meters. What is the length?

EXAMPLE 9.32

The perimeter of a rectangular swimming pool is 150 feet. The length is 15 feet more than the width. Find the length and width.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



Step 2. **Identify** what you are looking for. — the length and width of the pool

Step 3. **Name.** Choose a variable to represent it. The length is 15 feet more than the width.

Let W = width

$W + 15$ = length

Step 4. **Translate.** Write the appropriate formula and substitute.

$$\underbrace{P}_{150} \quad \underbrace{=}_{=} \quad \underbrace{2L}_{2(w+15)} \quad \underbrace{+}_{+} \quad \underbrace{2W}_{2w}$$

Step 5. **Solve** the equation.

$150 = 2w + 30 + 2w$

$150 = 4w + 30$

$120 = 4w$

$30 = w$ the width of the pool

$w + 15$ the length of the pool

$30 + 15$

45

Step 6. **Check:**

$p = 2L + 2W$

$150 \stackrel{?}{=} 2(45) + 2(30)$

$150 = 150$

Step 7. **Answer** the question.	The length of the pool is 45 feet and the width is 30 feet.

TRY IT 9.63 The perimeter of a rectangular swimming pool is 200 feet. The length is 40 feet more than the width. Find the length and width.

TRY IT 9.64 The length of a rectangular garden is 30 yards more than the width. The perimeter is 300 yards. Find the length and width.

Use the Properties of Triangles

We now know how to find the area of a rectangle. We can use this fact to help us visualize the formula for the area of a triangle. In the rectangle in Figure 9.20, we've labeled the length b and the width h, so it's area is bh.

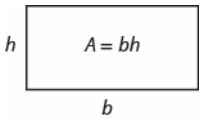


Figure 9.21 The area of a rectangle is the base, b, times the height, h.

We can divide this rectangle into two **congruent** triangles (Figure 9.22). Triangles that are congruent have identical side lengths and angles, and so their areas are equal. The area of each triangle is one-half the area of the rectangle, or $\frac{1}{2}bh$. This example helps us see why the formula for the area of a triangle is $A = \frac{1}{2}bh$.

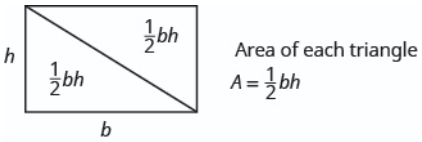


Figure 9.22 A rectangle can be divided into two triangles of equal area. The area of each triangle is one-half the area of the rectangle.

The formula for the area of a triangle is $A = \frac{1}{2}bh$, where b is the base and h is the height.

To find the area of the triangle, you need to know its base and height. The base is the length of one side of the triangle, usually the side at the bottom. The height is the length of the line that connects the base to the opposite vertex, and makes a $90°$ angle with the base. Figure 9.23 shows three triangles with the base and height of each marked.

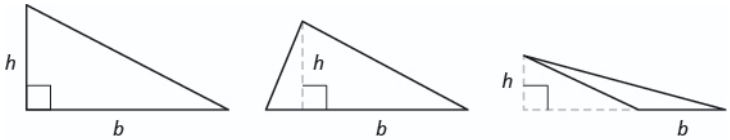


Figure 9.23 The height h of a triangle is the length of a line segment that connects the the base to the opposite vertex and makes a $90°$ angle with the base.

Triangle Properties

For any triangle ΔABC, the sum of the measures of the angles is $180°$.

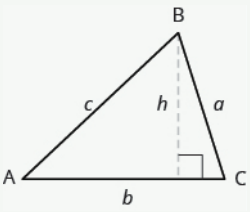
$$m\angle A + m\angle B + m\angle C = 180°$$

The perimeter of a triangle is the sum of the lengths of the sides.

$$P = a + b + c$$

The area of a triangle is one-half the base, b, times the height, h.

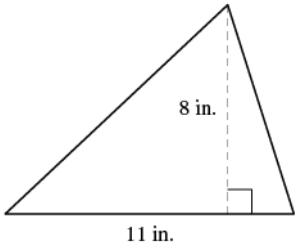
$$A = \frac{1}{2}bh$$



EXAMPLE 9.33

Find the area of a triangle whose base is 11 inches and whose height is 8 inches.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the area of the triangle
Step 3. **Name.** Choose a variable to represent it.	let A = area of the triangle
Step 4.**Translate.** Write the appropriate formula. Substitute.	$A = \frac{1}{2} \cdot b \cdot h$ $A = \frac{1}{2} \cdot 11 \cdot 8$
Step 5. **Solve** the equation.	$A = 44$ square inches
Step 6. **Check:** $A = \frac{1}{2}bh$ $44 \stackrel{?}{=} \frac{1}{2}(11)8$ $44 = 44$ ✓	
Step 7. **Answer** the question.	The area is 44 square inches.

TRY IT 9.65 Find the area of a triangle with base 13 inches and height 2 inches.

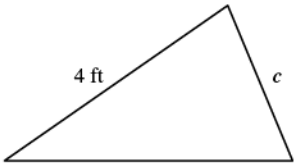
TRY IT 9.66 Find the area of a triangle with base 14 inches and height 7 inches.

EXAMPLE 9.34

The perimeter of a triangular garden is 24 feet. The lengths of two sides are 4 feet and 9 feet. How long is the third side?

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



$P = 24$ ft

Step 2. **Identify** what you are looking for.	length of the third side of a triangle
Step 3. **Name.** Choose a variable to represent it.	Let c = the third side
Step 4.**Translate.** Write the appropriate formula. Substitute in the given information.	$\underbrace{P}_{24} = \underbrace{a}_{4} + \underbrace{b}_{9} + \underbrace{c}_{c}$
Step 5. **Solve** the equation.	$24 = 13 + c$ $11 = c$
Step 6. **Check:** $P = a + b + c$ $24 \stackrel{?}{=} 4 + 9 + 11$ $24 = 24$ ✓	
Step 7. **Answer** the question.	The third side is 11 feet long.

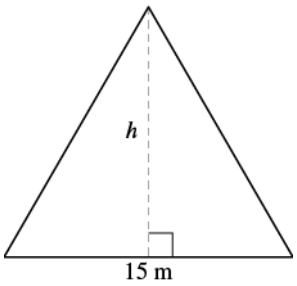
TRY IT 9.67 The perimeter of a triangular garden is 48 feet. The lengths of two sides are 18 feet and 22 feet. How long is the third side?

TRY IT 9.68 The lengths of two sides of a triangular window are 7 feet and 5 feet. The perimeter is 18 feet. How long is the third side?

EXAMPLE 9.35

The area of a triangular church window is 90 square meters. The base of the window is 15 meters. What is the window's height?

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	height of a triangle
Step 3. **Name.** Choose a variable to represent it.	Let h = the height
Step 4.**Translate.** Write the appropriate formula. Substitute in the given information.	$A = \frac{1}{2} \cdot b \cdot h$ $90 = \frac{1}{2} \cdot 15 \cdot h$
Step 5. **Solve** the equation.	$90 = \frac{15}{2}h$ $12 = h$
Step 6. **Check:** $A = \frac{1}{2}bh$ $90 \stackrel{?}{=} \frac{1}{2} \cdot 15 \cdot 12$ $90 = 90$ ✓	
Step 7. **Answer** the question.	The height of the triangle is 12 meters.

TRY IT 9.69 The area of a triangular painting is 126 square inches. The base is 18 inches. What is the height?

TRY IT 9.70 A triangular tent door has an area of 15 square feet. The height is 5 feet. What is the base?

Isosceles and Equilateral Triangles

Besides the right triangle, some other triangles have special names. A triangle with two sides of equal length is called an **isosceles triangle**. A triangle that has three sides of equal length is called an **equilateral triangle**. Figure 9.24 shows both types of triangles.

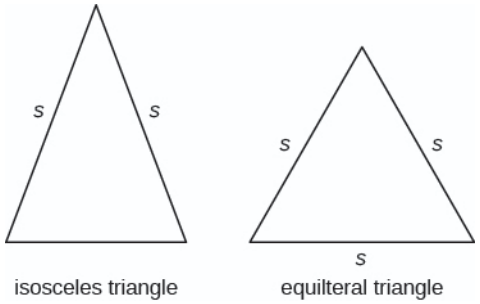


Figure 9.24 In an isosceles triangle, two sides have the same length, and the third side is the base. In an equilateral triangle, all three sides have the same length.

Isosceles and Equilateral Triangles

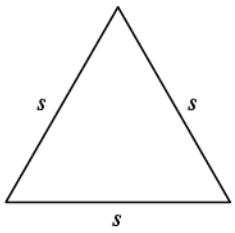
An **isosceles** triangle has two sides the same length.

An **equilateral** triangle has three sides of equal length.

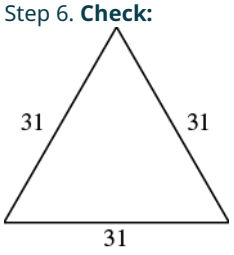
EXAMPLE 9.36

The perimeter of an equilateral triangle is 93 inches. Find the length of each side.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	 Perimeter = 93 in.
Step 2. **Identify** what you are looking for.	length of the sides of an equilateral triangle
Step 3. **Name.** Choose a variable to represent it.	Let s = length of each side
Step 4.**Translate.** Write the appropriate formula. Substitute.	$\underbrace{P}_{93} \underbrace{=}_{=} \underbrace{a}_{s} \underbrace{+}_{+} \underbrace{b}_{s} \underbrace{+}_{+} \underbrace{c}_{s}$
Step 5. **Solve** the equation.	$93 = 3s$ $31 = s$

Step 6. **Check:**



$93 \stackrel{?}{=} 31 + 31 + 31$

$93 = 93$ ✓

Step 7. **Answer** the question.	Each side is 31 inches.

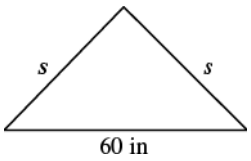
TRY IT 9.71 Find the length of each side of an equilateral triangle with perimeter 39 inches.

TRY IT 9.72 Find the length of each side of an equilateral triangle with perimeter 51 centimeters.

EXAMPLE 9.37

Arianna has 156 inches of beading to use as trim around a scarf. The scarf will be an isosceles triangle with a base of 60 inches. How long can she make the two equal sides?

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	 P = 156 in.
Step 2. **Identify** what you are looking for.	the lengths of the two equal sides
Step 3. **Name.** Choose a variable to represent it.	Let s = the length of each side
Step 4.**Translate.** Write the appropriate formula. Substitute in the given information.	$\underbrace{P}_{156} \underbrace{=}_{=} \underbrace{a}_{s} \underbrace{+}_{+} \underbrace{b}_{60} \underbrace{+}_{+} \underbrace{c}_{s}$
Step 5. **Solve** the equation.	$156 = 2s + 60$ $96 = 2s$ $48 = s$

Step 6. **Check:**

$p = a + b + c$

$156 \stackrel{?}{=} 48 + 60 + 48$

$156 = 156$ ✓

Step 7. **Answer** the question.	Arianna can make each of the two equal sides 48 inches long.

TRY IT 9.73 A backyard deck is in the shape of an isosceles triangle with a base of 20 feet. The perimeter of the deck is 48 feet. How long is each of the equal sides of the deck?

TRY IT 9.74 A boat's sail is an isosceles triangle with base of 8 meters. The perimeter is 22 meters. How long is each of the equal sides of the sail?

Use the Properties of Trapezoids

A **trapezoid** is four-sided figure, a *quadrilateral*, with two sides that are parallel and two sides that are not. The parallel sides are called the bases. We call the length of the smaller base b, and the length of the bigger base B. The height, h, of a trapezoid is the distance between the two bases as shown in Figure 9.25.

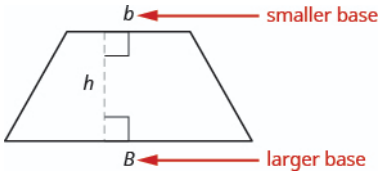


Figure 9.25 A trapezoid has a larger base, B, and a smaller base, b. The height h is the distance between the bases.

The formula for the area of a trapezoid is:

$$\text{Area}_{\text{trapezoid}} = \frac{1}{2}h(b + B)$$

Splitting the trapezoid into two triangles may help us understand the formula. The area of the trapezoid is the sum of the areas of the two triangles. See Figure 9.26.

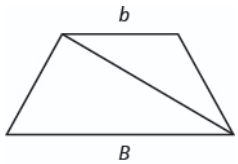


Figure 9.26 Splitting a trapezoid into two triangles may help you understand the formula for its area.

The height of the trapezoid is also the height of each of the two triangles. See Figure 9.27.

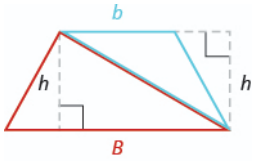


Figure 9.27

The formula for the area of a trapezoid is

$\text{Area}_{\text{trapezoid}} = \frac{1}{2}h(b + B)$

If we distribute, we get,

$$\text{Area}_{trapezoid} = \frac{1}{2}bh + \frac{1}{2}Bh$$

$$\text{Area}_{trapezoid} = A_{blue\Delta} + A_{red\Delta}$$

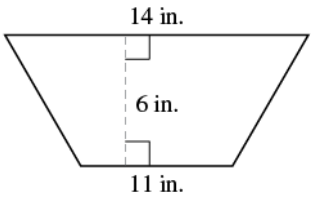
Properties of Trapezoids

- A trapezoid has four sides. See Figure 9.25.
- Two of its sides are parallel and two sides are not.
- The area, A, of a trapezoid is $A = \frac{1}{2}h(b + B)$.

EXAMPLE 9.38

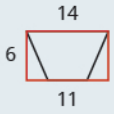
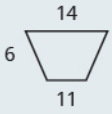
Find the area of a trapezoid whose height is 6 inches and whose bases are 14 and 11 inches.

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.	
Step 2. **Identify** what you are looking for.	the area of the trapezoid
Step 3. **Name.** Choose a variable to represent it.	Let A = the area
Step 4.**Translate.** Write the appropriate formula. Substitute.	$A = \frac{1}{2} \cdot h \cdot (b + B)$ $A = \frac{1}{2} \cdot 6 \cdot (11 + 14)$
Step 5. **Solve** the equation.	$A = \frac{1}{2} \cdot 6(25)$ $A = 3(25)$ $A = 75$ square inches
Step 6. **Check:** Is this answer reasonable?	

If we draw a rectangle around the trapezoid that has the same big base B and a height h, its area should be greater than that of the trapezoid.

If we draw a rectangle inside the trapezoid that has the same little base b and a height h, its area should be smaller than that of the trapezoid.

		
$A_{rectangle} = bh$	$A_{trapezoid} = \frac{1}{2}h(b + B)$	$A_{rectangle} = bh$
$A_{rectangle} = 14 \cdot 6$	$A_{trapezoid} = \frac{1}{2} \cdot 6(11 + 14)$	$A_{rectangle} = 11 \cdot 6$
$A_{rectangle} = 84$ sq. in.	$A_{trapezoid} = 75$ sq. in.	$A_{rectangle} = 66$ sq. in.

The area of the larger rectangle is 84 square inches and the area of the smaller rectangle is 66 square inches. So it makes sense that the area of the trapezoid is between 84 and 66 square inches

Step 7. **Answer** the question. The area of the trapezoid is 75 square inches.

TRY IT 9.75 The height of a trapezoid is 14 yards and the bases are 7 and 16 yards. What is the area?

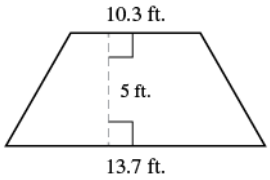
TRY IT 9.76 The height of a trapezoid is 18 centimeters and the bases are 17 and 8 centimeters. What is the area?

EXAMPLE 9.39

Find the area of a trapezoid whose height is 5 feet and whose bases are 10.3 and 13.7 feet.

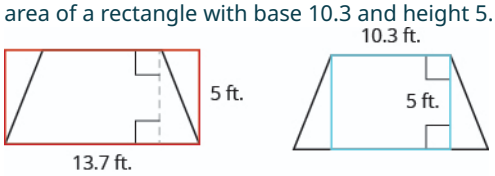
Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



Step 2. **Identify** what you are looking for.	the area of the trapezoid
Step 3. **Name.** Choose a variable to represent it.	Let A = the area
Step 4.**Translate.** Write the appropriate formula. Substitute.	$\underbrace{A}\ \underbrace{=}\ \underbrace{\frac{1}{2}}\ \underbrace{\cdot}\ \underbrace{h}\ \underbrace{\cdot}\ \underbrace{(b + B)}$ $A = \frac{1}{2} \cdot 5 \cdot (10.3 + 13.7)$
Step 5. **Solve** the equation.	$A = \frac{1}{2} \cdot 5(24)$ $A = 12 \cdot 5$ $A = 60$ square feet

Step 6. **Check:** Is this answer reasonable?
The area of the trapezoid should be less than the area of a rectangle with base 13.7 and height 5, but more than the area of a rectangle with base 10.3 and height 5.



$A_{rectangle} > A_{trapezoid} > A_{rectangle}$
68.5 60 51.5

Step 7. **Answer** the question.	The area of the trapezoid is 60 square feet.

TRY IT 9.77 The height of a trapezoid is 7 centimeters and the bases are 4.6 and 7.4 centimeters. What is the area?

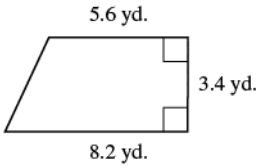
TRY IT 9.78 The height of a trapezoid is 9 meters and the bases are 6.2 and 7.8 meters. What is the area?

EXAMPLE 9.40

Vinny has a garden that is shaped like a trapezoid. The trapezoid has a height of 3.4 yards and the bases are 8.2 and 5.6 yards. How many square yards will be available to plant?

Solution

Step 1. **Read** the problem. Draw the figure and label it with the given information.



Step 2. **Identify** what you are looking for.	the area of a trapezoid
Step 3. **Name.** Choose a variable to represent it.	Let A = the area
Step 4.**Translate.** Write the appropriate formula. Substitute.	$\underbrace{A}_{}\ =\ \underbrace{\frac{1}{2}}_{}\ \cdot\ \underbrace{h}_{}\ \cdot\ \underbrace{(b+B)}_{}$ $A = \frac{1}{2} \cdot 3.4 \cdot (5.6 + 8.2)$
Step 5. **Solve** the equation.	$A = \frac{1}{2}(3.4)\,(13.8)$ $A = 23.46$ square yards

Step 6. **Check:** Is this answer reasonable?
Yes. The area of the trapezoid is less than the area of a rectangle with a base of 8.2 yd and height 3.4 yd, but more than the area of a rectangle with base 5.6 yd and height 3.4 yd.

$A_{rectangle} = Bh$ $= (8.2)(3.4)$ $= 27.88\ yd^2$	$A_{trapezoid} = \frac{1}{2}(3.4\ yd)(5.6 = 8.2)$ $= 23.46\ yd^2$	$A_{rectangle} = bh$ $= (5.6)(3.4)$ $= 19.04\ yd^2$
$A_{rectangle} > A_{trapezoid} > A_{rectangle}$ 27.88 23.46 19.04		

Step 7. **Answer** the question.	Vinny has 23.46 square yards in which he can plant.

TRY IT 9.79 Lin wants to sod his lawn, which is shaped like a trapezoid. The bases are 10.8 yards and 6.7 yards, and the height is 4.6 yards. How many square yards of sod does he need?

TRY IT 9.80 Kira wants cover his patio with concrete pavers. If the patio is shaped like a trapezoid whose bases are 18 feet and 14 feet and whose height is 15 feet, how many square feet of pavers will he need?

LINKS TO LITERACY

The Links to Literacy activity *Spaghetti and Meatballs for All* will provide you with another view of the topics covered in this section."

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES
Perimeter of a Rectangle (https://www.openstax.org/l/24perirect)
Area of a Rectangle (https://www.openstax.org/l/24arearect)
Perimeter and Area Formulas (https://www.openstax.org/l/24periareaform)
Area of a Triangle (https://www.openstax.org/l/24areatri)
Area of a Triangle with Fractions (https://www.openstax.org/l/24areatrifract)
Area of a Trapezoid (https://www.openstax.org/l/24areatrap)

SECTION 9.4 EXERCISES

Practice Makes Perfect

Understand Linear, Square, and Cubic Measure

In the following exercises, determine whether you would measure each item using linear, square, or cubic units.

129. amount of water in a fish tank

130. length of dental floss

131. living area of an apartment

132. floor space of a bathroom tile

133. height of a doorway

134. capacity of a truck trailer

In the following exercises, find the ⓐ perimeter and ⓑ area of each figure. Assume each side of the square is 1 cm.

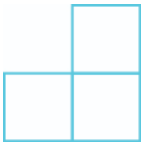
135.



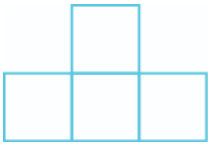
136.



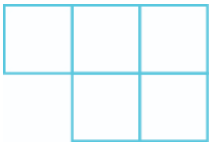
137.



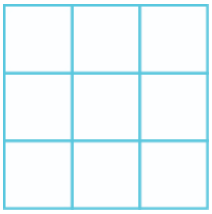
138.



139.



140.



Use the Properties of Rectangles

In the following exercises, find the ⓐ perimeter and ⓑ area of each rectangle.

141. The length of a rectangle is 85 feet and the width is 45 feet.

142. The length of a rectangle is 26 inches and the width is 58 inches.

143. A rectangular room is 15 feet wide by 14 feet long.

144. A driveway is in the shape of a rectangle 20 feet wide by 35 feet long.

In the following exercises, solve.

145. Find the length of a rectangle with perimeter 124 inches and width 38 inches.

146. Find the length of a rectangle with perimeter 20.2 yards and width of 7.8 yards.

147. Find the width of a rectangle with perimeter 92 meters and length 19 meters.

148. Find the width of a rectangle with perimeter 16.2 meters and length 3.2 meters.

149. The area of a rectangle is 414 square meters. The length is 18 meters. What is the width?

150. The area of a rectangle is 782 square centimeters. The width is 17 centimeters. What is the length?

151. The length of a rectangle is 9 inches more than the width. The perimeter is 46 inches. Find the length and the width.

152. The width of a rectangle is 8 inches more than the length. The perimeter is 52 inches. Find the length and the width.

153. The perimeter of a rectangle is 58 meters. The width of the rectangle is 5 meters less than the length. Find the length and the width of the rectangle.

154. The perimeter of a rectangle is 62 feet. The width is 7 feet less than the length. Find the length and the width.

155. The width of the rectangle is 0.7 meters less than the length. The perimeter of a rectangle is 52.6 meters. Find the dimensions of the rectangle.

156. The length of the rectangle is 1.1 meters less than the width. The perimeter of a rectangle is 49.4 meters. Find the dimensions of the rectangle.

157. The perimeter of a rectangle of 150 feet. The length of the rectangle is twice the width. Find the length and width of the rectangle.

158. The length of a rectangle is three times the width. The perimeter is 72 feet. Find the length and width of the rectangle.

159. The length of a rectangle is 3 meters less than twice the width. The perimeter is 36 meters. Find the length and width.

160. The length of a rectangle is 5 inches more than twice the width. The perimeter is 34 inches. Find the length and width.

161. The width of a rectangular window is 24 inches. The area is 624 square inches. What is the length?

162. The length of a rectangular poster is 28 inches. The area is 1316 square inches. What is the width?

163. The area of a rectangular roof is 2310 square meters. The length is 42 meters. What is the width?

164. The area of a rectangular tarp is 132 square feet. The width is 12 feet. What is the length?

165. The perimeter of a rectangular courtyard is 160 feet. The length is 10 feet more than the width. Find the length and the width.

166. The perimeter of a rectangular painting is 306 centimeters. The length is 17 centimeters more than the width. Find the length and the width.

167. The width of a rectangular window is 40 inches less than the height. The perimeter of the doorway is 224 inches. Find the length and the width.

168. The width of a rectangular playground is 7 meters less than the length. The perimeter of the playground is 46 meters. Find the length and the width.

Use the Properties of Triangles

In the following exercises, solve using the properties of triangles.

169. Find the area of a triangle with base 12 inches and height 5 inches.

170. Find the area of a triangle with base 45 centimeters and height 30 centimeters.

171. Find the area of a triangle with base 8.3 meters and height 6.1 meters.

172. Find the area of a triangle with base 24.2 feet and height 20.5 feet.

173. A triangular flag has base of 1 foot and height of 1.5 feet. What is its area?

174. A triangular window has base of 8 feet and height of 6 feet. What is its area?

175. If a triangle has sides of 6 feet and 9 feet and the perimeter is 23 feet, how long is the third side?

176. If a triangle has sides of 14 centimeters and 18 centimeters and the perimeter is 49 centimeters, how long is the third side?

177. What is the base of a triangle with an area of 207 square inches and height of 18 inches?

178. What is the height of a triangle with an area of 893 square inches and base of 38 inches?

179. The perimeter of a triangular reflecting pool is 36 yards. The lengths of two sides are 10 yards and 15 yards. How long is the third side?

180. A triangular courtyard has perimeter of 120 meters. The lengths of two sides are 30 meters and 50 meters. How long is the third side?

181. An isosceles triangle has a base of 20 centimeters. If the perimeter is 76 centimeters, find the length of each of the other sides.

182. An isosceles triangle has a base of 25 inches. If the perimeter is 95 inches, find the length of each of the other sides.

183. Find the length of each side of an equilateral triangle with a perimeter of 51 yards.

184. Find the length of each side of an equilateral triangle with a perimeter of 54 meters.

185. The perimeter of an equilateral triangle is 18 meters. Find the length of each side.

186. The perimeter of an equilateral triangle is 42 miles. Find the length of each side.

187. The perimeter of an isosceles triangle is 42 feet. The length of the shortest side is 12 feet. Find the length of the other two sides.

188. The perimeter of an isosceles triangle is 83 inches. The length of the shortest side is 24 inches. Find the length of the other two sides.

189. A dish is in the shape of an equilateral triangle. Each side is 8 inches long. Find the perimeter.

190. A floor tile is in the shape of an equilateral triangle. Each side is 1.5 feet long. Find the perimeter.

191. A road sign in the shape of an isosceles triangle has a base of 36 inches. If the perimeter is 91 inches, find the length of each of the other sides.

192. A scarf in the shape of an isosceles triangle has a base of 0.75 meters. If the perimeter is 2 meters, find the length of each of the other sides.

193. The perimeter of a triangle is 39 feet. One side of the triangle is 1 foot longer than the second side. The third side is 2 feet longer than the second side. Find the length of each side.

194. The perimeter of a triangle is 35 feet. One side of the triangle is 5 feet longer than the second side. The third side is 3 feet longer than the second side. Find the length of each side.

195. One side of a triangle is twice the smallest side. The third side is 5 feet more than the shortest side. The perimeter is 17 feet. Find the lengths of all three sides.

196. One side of a triangle is three times the smallest side. The third side is 3 feet more than the shortest side. The perimeter is 13 feet. Find the lengths of all three sides.

Use the Properties of Trapezoids

In the following exercises, solve using the properties of trapezoids.

197. The height of a trapezoid is 12 feet and the bases are 9 and 15 feet. What is the area?

198. The height of a trapezoid is 24 yards and the bases are 18 and 30 yards. What is the area?

199. Find the area of a trapezoid with a height of 51 meters and bases of 43 and 67 meters.

200. Find the area of a trapezoid with a height of 62 inches and bases of 58 and 75 inches.

201. The height of a trapezoid is 15 centimeters and the bases are 12.5 and 18.3 centimeters. What is the area?

202. The height of a trapezoid is 48 feet and the bases are 38.6 and 60.2 feet. What is the area?

203. Find the area of a trapezoid with a height of 4.2 meters and bases of 8.1 and 5.5 meters.

204. Find the area of a trapezoid with a height of 32.5 centimeters and bases of 54.6 and 41.4 centimeters.

205. Laurel is making a banner shaped like a trapezoid. The height of the banner is 3 feet and the bases are 4 and 5 feet. What is the area of the banner?

206. Niko wants to tile the floor of his bathroom. The floor is shaped like a trapezoid with width 5 feet and lengths 5 feet and 8 feet. What is the area of the floor?

207. Theresa needs a new top for her kitchen counter. The counter is shaped like a trapezoid with width 18.5 inches and lengths 62 and 50 inches. What is the area of the counter?

208. Elena is knitting a scarf. The scarf will be shaped like a trapezoid with width 8 inches and lengths 48.2 inches and 56.2 inches. What is the area of the scarf?

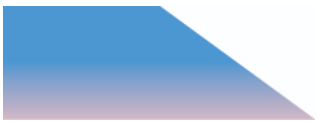
Everyday Math

209. **Fence** Jose just removed the children's playset from his back yard to make room for a rectangular garden. He wants to put a fence around the garden to keep out the dog. He has a 50 foot roll of fence in his garage that he plans to use. To fit in the backyard, the width of the garden must be 10 feet. How long can he make the other side if he wants to use the entire roll of fence?

210. **Gardening** Lupita wants to fence in her tomato garden. The garden is rectangular and the length is twice the width. It will take 48 feet of fencing to enclose the garden. Find the length and width of her garden.

211. **Fence** Christa wants to put a fence around her triangular flowerbed. The sides of the flowerbed are 6 feet, 8 feet, and 10 feet. The fence costs $\$10$ per foot. How much will it cost for Christa to fence in her flowerbed?

212. **Painting** Caleb wants to paint one wall of his attic. The wall is shaped like a trapezoid with height 8 feet and bases 20 feet and 12 feet. The cost of the painting one square foot of wall is about $\$0.05$. About how much will it cost for Caleb to paint the attic wall?



Writing Exercises

213. If you need to put tile on your kitchen floor, do you need to know the perimeter or the area of the kitchen? Explain your reasoning.

214. If you need to put a fence around your backyard, do you need to know the perimeter or the area of the backyard? Explain your reasoning.

215. Look at the two figures.



ⓐ Which figure looks like it has the larger area? Which looks like it has the larger perimeter?
ⓑ Now calculate the area and perimeter of each figure. Which has the larger area? Which has the larger perimeter?

216. The length of a rectangle is 5 feet more than the width. The area is 50 square feet. Find the length and the width.
ⓐ Write the equation you would use to solve the problem.
ⓑ Why can't you solve this equation with the methods you learned in the previous chapter?